The first law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a consistent sign convention in an energy balance.

SUCCESS CRITERIA - I can...

1. State whether heat enters or leaves.
2. Distinguish work done by the system from work done on it.
3. Apply ΔU=Q-W with the stated signs.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 0.3 kg vapor releases 100000 J while condensing at constant temperature. Find latent heat and signed Q for the sample.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\Delta U = Q - W_{by}$.
Q is positive into the system; W_{by} is positive when the system does work on its surroundings.

Worked reasoning: A system receives 90 J of heat and does 30 J of work on its surroundings. With ΔU=Q-W, internal energy increases by 60 J.

Your turn: Draw labeled arrows crossing a system boundary and mark the 60 J stored increase.

Draw and label all energy transfers across the boundary.

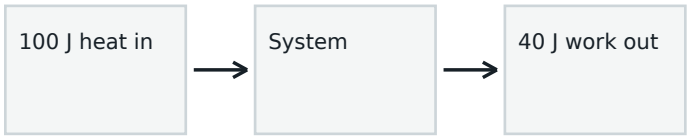
Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and interpret the model

A system absorbs 100 J of heat and does 40 J of work. Draw the energy arrows and find its internal-energy change.



Marzano check - circle: ① ② ③

04 Apply the physics

A gas loses 50 J of heat while a piston does 60 J of work on it. Label both transfers on the system boundary. Use $\Delta U = Q - W_{by}$, with work done by the gas positive, to calculate ΔU.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

05 Explain your reasoning

Why does the phrase “work is 60 J” need a direction or sign convention?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 4 kg cart at 12 m/s hits an identical cart at rest. Final velocities are 5 and 7 m/s. Check momentum and energy; classify the collision.
Show the energy changes and transfers; label units.

Initial | **Transfers** | **Final**



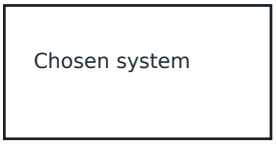
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

A process has ΔU=-15 J and W_{by}=25 J. Find Q and say whether heat enters or leaves. Label energy entering, leaving or stored on the system model.

Draw and label all energy transfers across the boundary.



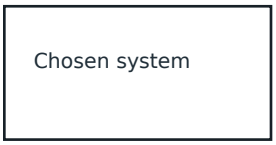
Stored energy change: _____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A system loses 20 J of heat and has 50 J of work done on it. Using work-by-system as positive, find Q, W and ΔU. Then reverse both transfers. Label energy entering, leaving or stored on the system model.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3